Jayadratha Gayen

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Python and GenAI Engineer with 5+ years of professional experience building production AI systems, LLM-powered agents, and agentic workflows. Hands-on work with LangGraph, LangChain, MCP, RAG pipelines, and LLM APIs in real product environments. Published researcher in graph neural networks and temporal graph learning. Comfortable owning the full stack from prototype to production.

Technical Skills

Languages: Python, SQL, C++

GenAI / Agentic AI: LangGraph, LangChain, MCP, Agentic AI, LLM APIs, RAG, Tool Use, LLM Evaluation

ML / DL: PyTorch, PyG, TensorFlow, Transformers, HuggingFace, scikit-learn, Graph Neural Networks

Infrastructure: AWS, Docker, Git, MongoDB, Neo4j, Vector Databases, Knowledge Graphs

Work Experience

CloudAngles Inc. Feb 2026 – Present
AI Research Engineer | *Hyderabad, India*

- Built core GenAI components for Codebenders, an agentic coding assistant: designed repository indexing with semantic retrieval, context-management pipelines, and tool orchestration using LangGraph and LLM APIs for natural language code editing, debugging, and generation.
- Reduced AI token usage by **76.3%** and API costs by **64.2%** through in-session tool result caching and smarter tool loading, keeping the agent cost-efficient at scale.
- Improved agent response speed by **22.4%** by reworking context window management and session memory, so the agent picks up exactly where it left off after any interruption.
- Built automated **LLM evaluation pipelines** with custom benchmarks for code quality, task completion rate, and model reliability across real developer scenarios.
- Worked with MCP to standardize tool interfaces and reduce integration overhead across the agentic system.

Machine Learning Lab (MLL), IIIT Hyderabad Nov 2022 – Jul 2025
Research Associate | *Hyderabad, India*

- Designed and trained graph neural network models in Python and PyTorch for node classification with uncertainty management, applied to legal judgment prediction and disease diagnosis; published at **TMLR 2025**.
- Built abstention frameworks for continuous time dynamic graphs (CTDGs), improving AUC/AP by **10–15%** under class imbalance; published at **ASONAM 2025** and **KDD TGL Workshop 2025**.
- Developed end-to-end ML pipelines in Python using PyTorch and PyG across legal, medical, and biological graph datasets, covering training, evaluation, and visualization.
- Worked on 3D computer vision research for pose and motion transfer using graph convolution networks.

Google Summer of Code at INCF May 2025 – Sep 2025
Open Source Developer | *Remote*

- Built **DevoTG** in Python: a temporal graph neural network framework for modeling C. elegans neural development as evolving graphs; temporal approach achieved mean test AUC of **0.839 vs 0.577** for static methods.
- Designed dataset pipelines for spatiotemporal biological data and built interactive **3D visualizations**.

Vikram Solar Limited Jul 2018 – Dec 2019
Engineer | *Falta, West Bengal, India*

- Led a Six Sigma throughput improvement project, increasing production by **10%** through bottleneck analysis using DOE, Pareto, and Fishbone methods.
- Introduced RFID automation and SAP-based data analytics workflows to reduce manual errors in manufacturing.

Education

International Institute of Information Technology Hyderabad
M.Sc in Electronics and Communication Engineering by Research (CGPA: 8.29/10) 2022 – 2025

Indian Institute of Engineering Science and Technology Shibpur
Bachelor of Technology in Electrical Engineering (CGPA: 8.02/10) 2014 – 2018

Selected Publications

Node Classification With Reject Option (**TMLR 2025**)

- Proposed **NodeCwR**, a reject-option GNN framework that abstains on uncertain predictions to improve reliability in high-stakes domains like legal and medical applications.

Reducing Misclassification Risk in Dynamic Graph Neural Networks Through Abstention (**ASONAM 2025**)

- Introduced abstention mechanisms for temporal GNNs, improving AUC/AP by **10–15%** while handling class imbalance and critical misclassifications.

DevoTG: Temporal GNNs for Modeling C. elegans Developmental Connectomics (**arXiv 2026**)

- Built a temporal GNN framework for modeling neural development; mean test AUC of 0.839 vs 0.577 for static methods.

Selected Projects

Nyay Saheli: Conversational Legal Assistant (LLM + GraphRAG) 2025

- Built a multilingual legal AI assistant using **RAG** and **GraphRAG** for legal guidance and question answering in Indian languages.
- Developed OCR ingestion, embedding, hybrid search, reranking, and retrieval pipelines over legal knowledge bases using vector databases and knowledge graphs.
- National Finalist (Top 6/275 teams) at **BharatGen Hackathon** for this project.

Node Classification with Reject Option for Legal Judgement Prediction (NLP + GNN) 2024

- Built NodeCwR, a GNN framework with cost-based and coverage-based rejection for Legal Judgment Prediction, analyzing complex legal texts using NLP and graph-based representations; achieved 6–19% accuracy improvement on the ILDC dataset.
- Enhanced interpretability of model decisions using SHAP visualizations.

Multilingual News Similarity Detection (Transformers + NLP) 2022

- Built a multilingual document similarity system using Siamese Transformer architectures; best result with **multilingual DistilBERT** (PCC: 0.5683) in SemEval 2022 Task 8.

Leadership & Activities

- National Finalist (Top 6/275 teams), **BharatGen Hackathon** – GenAI multilingual legal assistant.
- Teaching Assistant, **3D Vision Summer School**, IIIT Hyderabad – conducted tutorials on graph neural networks and held doubt sessions for participants from across India.